Supplementary Materials for

Cascading Foreshocks, Aftershocks, and earthquake swarms in a Discrete Fault Network

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ETAS parameter inversion - Maximum Likelihood Estimation (MLE)

We conducted maximum likelihood estimation (MLE) with all the catalogs from our discrete fault simulations (figure 8). The MLE method finds the best fitting ETAS model by maximizing the following log-likelihood [Ogata, 1983],

$$\log L(\varphi) = \sum_i \lambda_\varphi(t_i \left| H_{t_i} \right.) - \int_0^T \lambda_\varphi(t_i \left| H_{t_i} \right.) dt \,, \tag{S1}$$

where H_t represents the occurrence of a series of earthquakes before time t, $\varphi = (\mu_0, K_0, \alpha, c, p)$ is the parameter vector tested, and [0, T] is the time window evaluated. We calculate the log-likelihood (equation S1) of each individual simulation catalog and maximize the sum of them all (all the simulations are run with the same model parameters). The MLE inversion is separately conducted using the catalog of three different periods of the simulation result (i) all duration, (ii) 500 days of foreshocks only, and (iii) 500 days of aftershocks only (table S1).

The inversion results (table S1) confirm well our finding (figure 12) that the foreshock and aftershock sequences have different productivity. In the MLE inversion (table S1), the productivity parameters, K and α, are considerably larger for foreshocks than for aftershocks. As we discussed in the main text, the higher productivity in foreshock sequences results from stronger interactions within the main fault zone, where the density of secondary faults is higher.

The MLE best fitting parameter set (table S1) somewhat differs from the parameters derived from fitting the stacked seismicity curves using the approach presented in the main text. The most distinct parameter is c. In MLE, the c is in the order of 10^{-7} day, which is several orders of magnitude smaller than the value derived from fitting the time evolution of seismicity shown in figure 12. The extremely small value of c obtained with the MLE method is presumably an artifact of the the under-resolved seismicity timing in the DFN simulation. To limit the data size, we record the simulation results only every 500 time steps, resulting in poorly resolved earthquake timing. For instance, those catalogs commonly have several events with an identical origin time. This issue seems to generate a bias in the MLE inversion, especially regarding the determination of c, which quantifies the duration of the seismic rate plateau in the early aftershock sequence. The parameters that define the Omori decay curve (c and p) are similar between foreshock and aftershock, presumably because this issue applies similarly to both sequences. By contrast, the productivity parameters (K and α) determined using the MLE method don't seem to be biased and are found to be significantly larger for the foreshocks than for the aftershocks.

Table S1. MLE parameter inversion result

	μ_0	c	p	K	α
All duration	0.040	4.44E-07	1.017	0.0084	0.776
Foreshock (500days)	0.049	5.10E-07	1.014	0.0114	0.810
Aftershock (500 days)	0.081	5.84E-07	1.084	0.0083	0.633

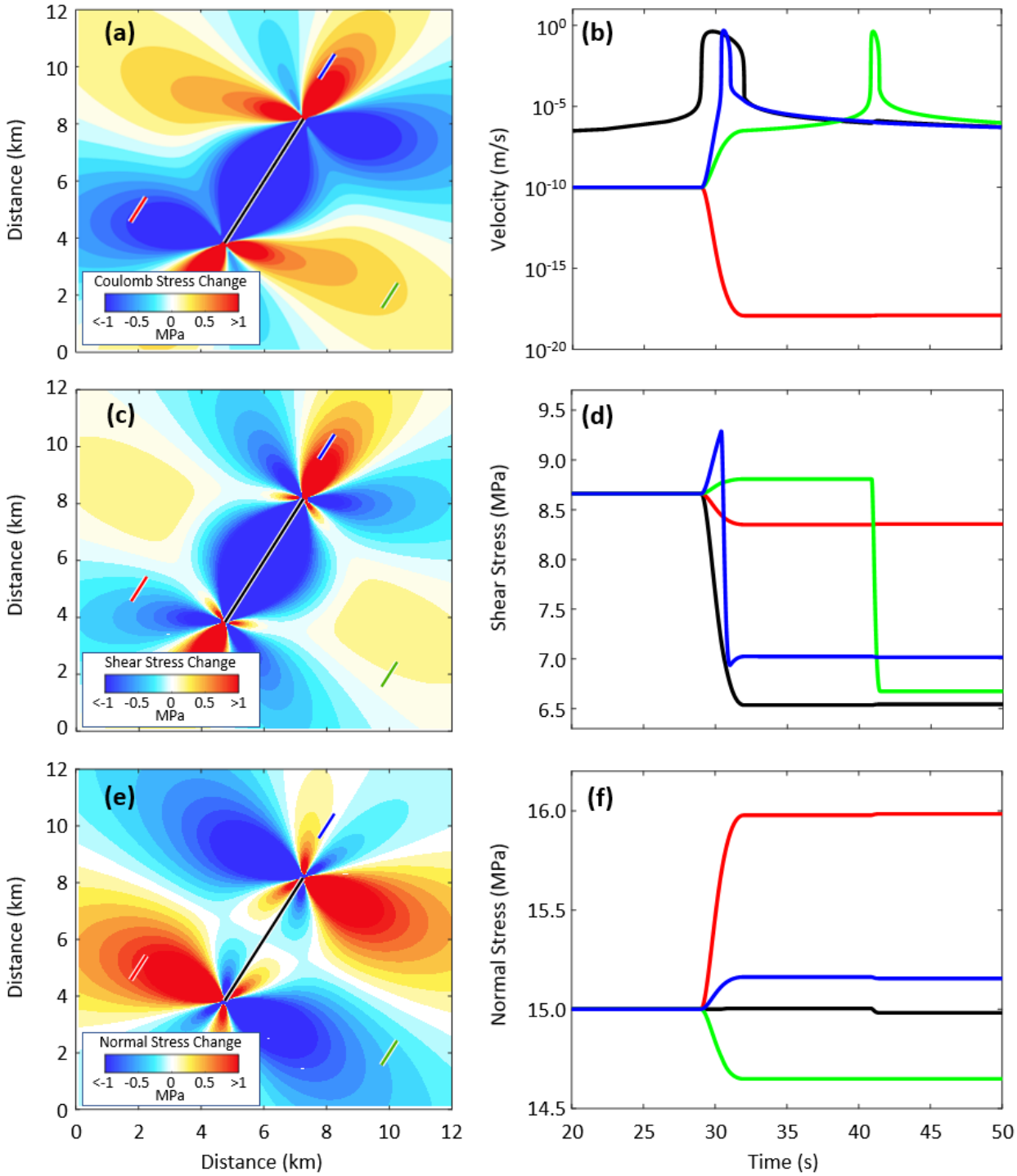

Figure S1. Identical to figure 3 but with slip law.

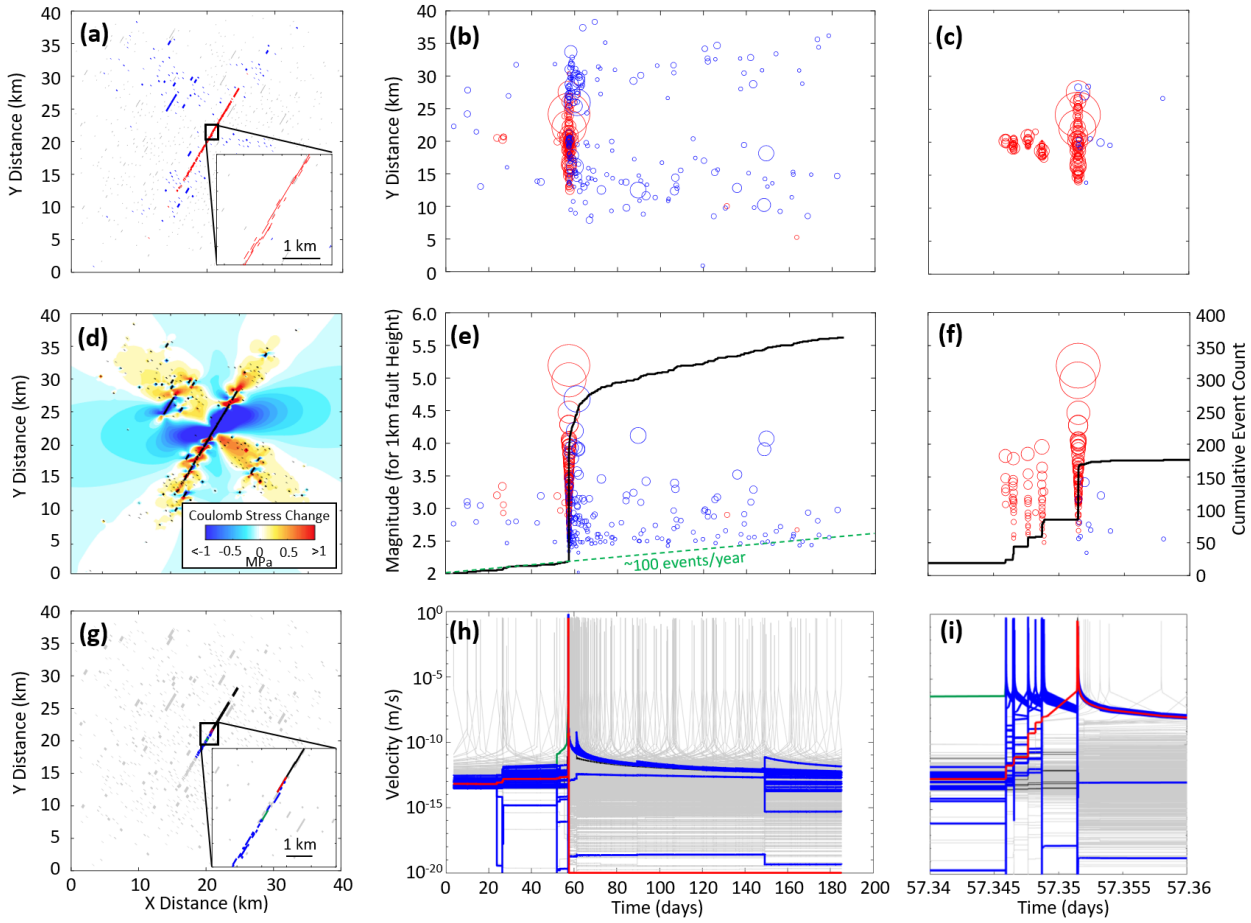

Figure S2. Identical to figure 5 but with slip law.

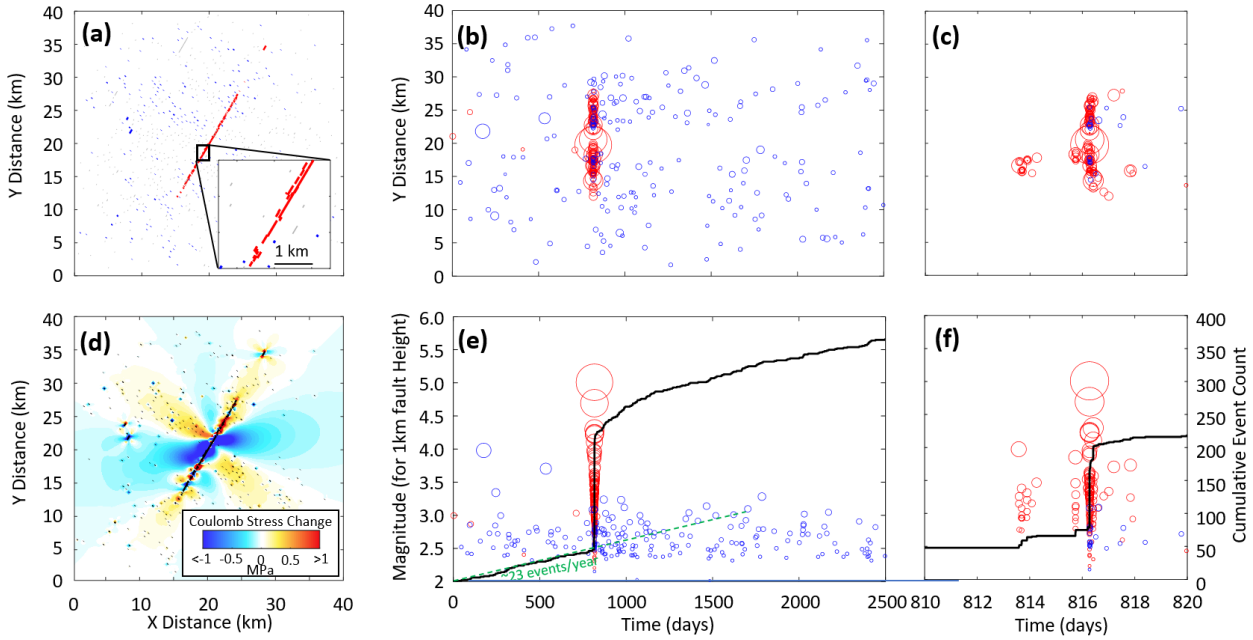

Figure S3. Simulation results with constant loading. The panel descriptions are identical to figure 5a-f of the main text. Fault distribution is randomly set with the identical method for the main simulation setup (complex fault zone simulation, section 3.3). We used constant shear stressing throughout the simulation period at 10kPa/year, which roughly corresponds with the inter-seismic loading rate of the south California San Andreas fault [Freed et al., 2007]. Initial velocity is randomly set by equation 17 (case with loading) to yield ~23 events/year when no interaction is assumed. Cascading foreshock is apparent a few days before the mainshock. The result is essentially similar to figure 5, which assumed no loading.

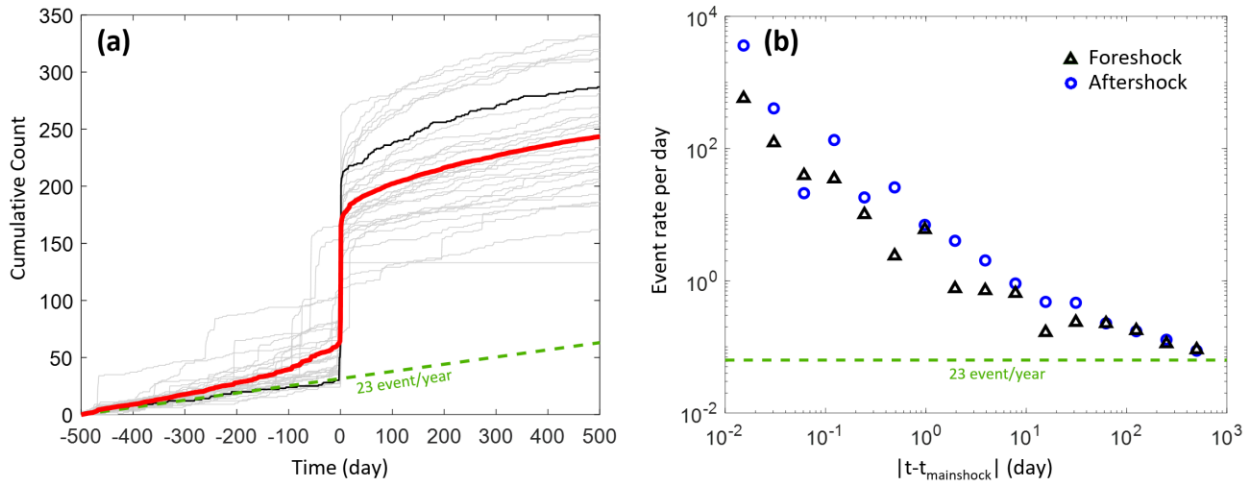

Figure S4. Stacked results from simulations with tectonic loading included (10kPa/year, cf Figure S3). We conducted 100 simulations. Among them, 31 simulations that have the mainshock after 500 days are selected and stacked. The simulation parameters are identical to figure S3 but with randomized geometry. **(a)**: Cumulative seismicity with time before and after the mainshock. Gray lines denote 31 simulation results that have mainshock occurring after 500 days. Figure S3 case is highlighted in the black line. The bold red line denotes the stacked average. **(b)**: Event rate vs. time form mainshock for both foreshock (black triangle) and aftershock (blue circles). The green dashed line denotes the seismicity rate in the case of no cascading interaction.

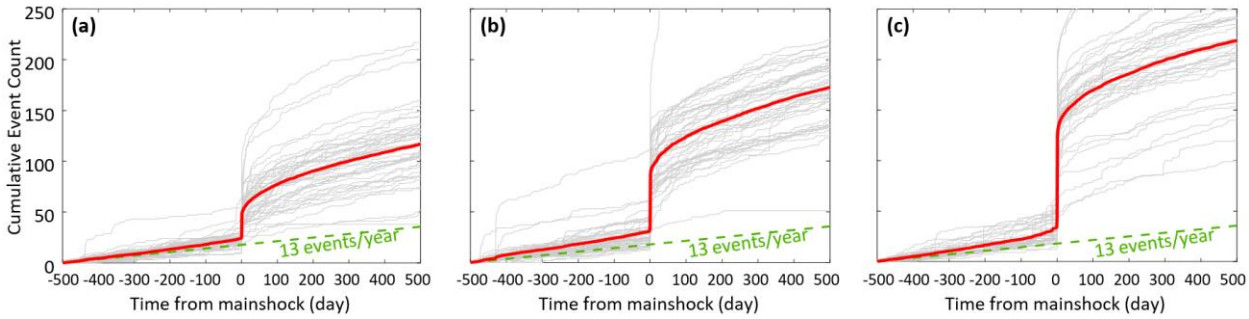

Figure S5. Cumulative seismicity with time before and after the mainshock for simulations with an initial number of faults in the fault zone of 50 (a), 100 (b), and 150 (c). The remaining number of faults after the application of the 'too close' criterion is 42 (a), 78 (b), and 108 (c). Gray lines show all simulation results, and bold red lines denote the stacked average.

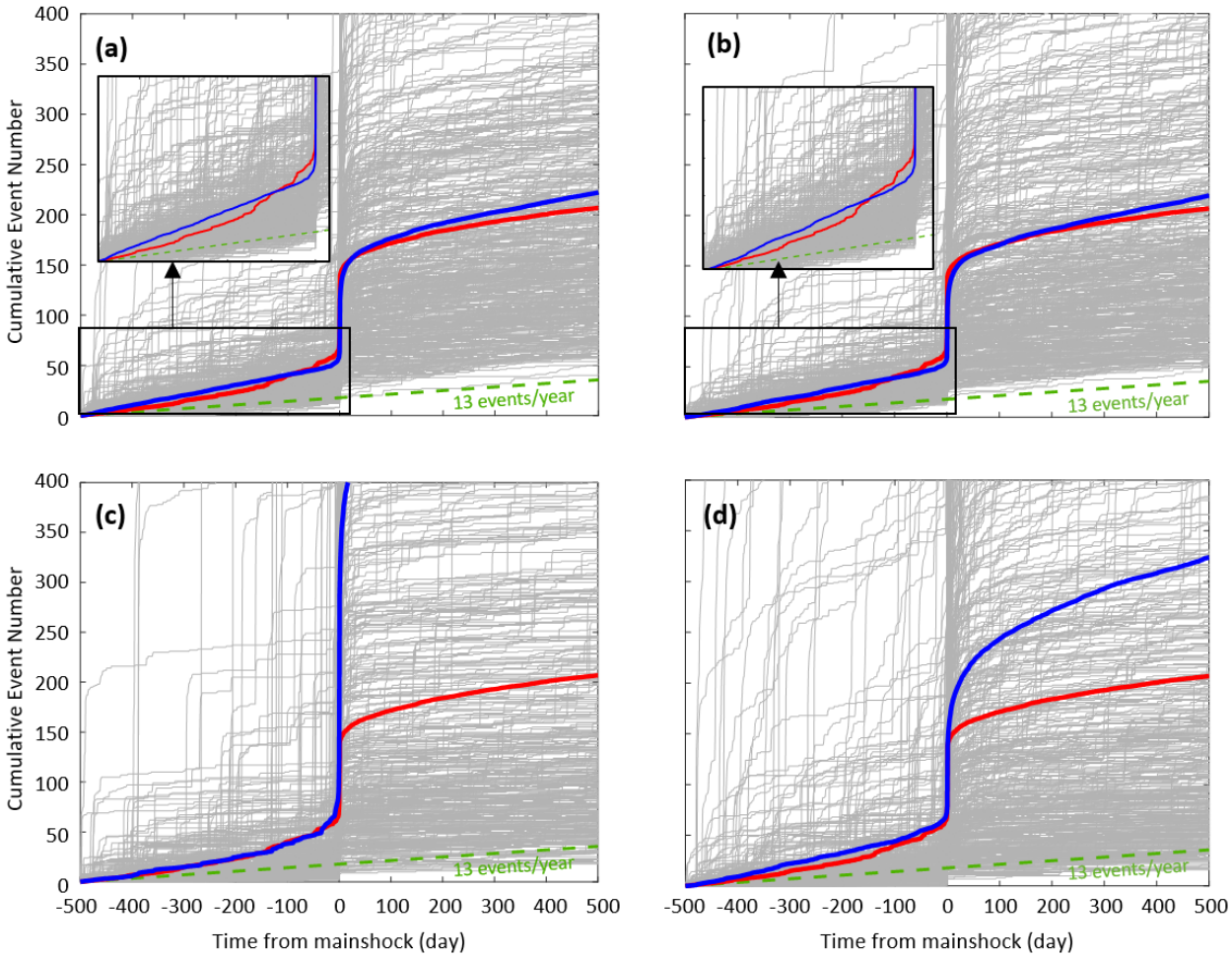

Figure S6. Additional attempts to fit foreshock (c, d) and aftershock (a, b) sequences in our discrete fault network model with ETAS. This figure corresponds to figure 12 but with different parameters. **(a)**: λ_0 = 13 event/year, c=0.028, θ= 0.3 (p=1.3), K=0.011 and α=1. **(b)**: λ_0 = 13 event/year, c=0.015, θ= 0.3 (p=1.3), K=0.011 and α=0.945. **(c)**: λ_0 = 6.5 event/year, c=0.001, θ= 0.3 (p=1.3), K=0.005 and α=1. **(d)**: λ_0 = 4.68 event/year, c=0.0005, θ= 0.1 (p=1.1), K=0.01 and α=0.93.